

Problem 2

In a survey, each participant is asked for their birth month. What is the minimum number of participants needed to guarantee that at least two were born in the same month?

Solution

There are 12 possible birth months. If 12 participants are chosen, it is still possible (in the worst case) that each was born in a different month, so no repetition is guaranteed. This matches the student's reasoning:

“It is related to the number of months. If there are 12 months, the minimum number of participants could be 12 without guaranteeing a match. To guarantee that at least two people have the same birth month, one more participant must be included.”

Therefore, with 13 participants, at least two must share a birth month.